

SAIQ, Oman

WMO: 412540

Lat: 23.076N

Lon: 57.643E

Elev: 6516

StdP: 11.55

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	38.4	41.0	-10.8	3.9	53.0	-3.4	5.8	54.4	22.6	51.4	20.0	52.5	4.3	20	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	89.3	56.7	87.6	56.7	86.1	56.6	64.8	76.9	63.5	76.8	62.2	77.0	8.2	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.5	100.6	73.1	58.5	93.6	73.1	56.5	86.8	71.6	33.7	76.7	32.5	76.2	31.4	77.1	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	17.1	14.6	DB	33.9	91.8	3.3	1.6	31.6	93.0	29.6	93.9	27.8	94.8	25.4	96.0	(4)
(5)			WB	25.8	67.5	2.2	1.6	24.2	68.6	23.0	69.6	21.7	70.4	20.2	71.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.9	51.2	54.2	60.2	66.4	73.1	78.6	78.9	77.1	73.1	66.3	57.9	52.9	(6)
(7)		DBStd	10.60	4.56	4.95	4.21	3.90	3.83	3.46	2.77	2.79	2.88	3.37	3.44	4.38	(7)
(8)		HDD50	84	40	19	1	0	0	0	0	0	0	0	0	24	(8)
(9)		HDD65	1531	427	302	156	31	0	0	0	0	0	26	213	376	(9)
(10)		CDD50	5883	79	138	315	492	717	858	896	841	692	505	237	113	(10)
(11)		CDD65	1855	0	1	6	72	253	408	431	376	242	66	0	0	(11)
(12)		CDH74	13675	0	1	8	178	1667	3884	3887	2708	1180	162	0	0	(12)
(13)		CDH80	3612	0	0	0	1	351	1282	1184	667	127	0	0	0	(13)
(14)	Wind	WSAvg	6.0	5.4	6.2	6.7	7.2	6.7	7.2	5.6	6.6	6.0	5.5	4.8	4.8	(14)
(15)	Precipitation	PrecAvg	7.8	0.5	0.8	1.1	0.9	0.5	0.4	1.1	0.9	0.6	0.4	0.4	0.4	(15)
(16)		PrecMax	25.4	3.2	4.4	13.1	2.9	2.8	2.4	5.1	3.1	3.4	2.2	2.4	1.7	(16)
(17)		PrecMin	3.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	4.3	0.9	1.0	2.5	1.0	0.6	0.6	1.2	0.8	0.7	0.6	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.3	70.7	75.4	79.5	90.6	92.0	90.5	89.0	84.6	79.4	71.1	67.9	(19)
(20)			MCWB	44.3	48.2	47.8	52.4	55.5	57.4	57.7	57.8	55.2	52.3	50.6	43.8	(20)
(21)		2%	DB	64.6	67.8	72.9	77.8	85.8	89.8	88.7	86.7	83.0	77.7	69.1	65.5	(21)
(22)			MCWB	44.7	46.8	47.9	51.6	54.3	56.2	57.8	57.9	55.3	52.1	50.3	44.3	(22)
(23)		5%	DB	62.3	65.7	70.9	76.2	83.7	88.0	87.2	85.2	81.6	75.8	67.6	63.6	(23)
(24)			MCWB	44.7	46.2	47.8	51.0	54.1	55.5	58.1	58.0	55.0	51.8	50.0	44.6	(24)
(25)		10%	DB	59.9	63.5	68.9	74.6	81.3	86.4	85.7	83.8	80.1	74.0	66.0	61.6	(25)
(26)			MCWB	43.8	46.0	47.6	50.6	53.6	55.0	57.9	57.9	54.6	51.3	49.1	45.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	53.9	56.5	62.6	61.8	63.5	66.6	66.2	62.0	59.8	58.0	52.9	(27)
(28)			MCD B	57.3	64.2	63.7	67.2	74.3	78.3	79.3	77.4	73.9	67.8	65.5	58.7	(28)
(29)		2%	WB	49.8	51.9	54.2	57.0	59.7	61.2	65.0	64.7	60.5	57.9	54.7	51.4	(29)
(30)			MCD B	56.5	60.3	62.6	66.3	73.2	77.9	78.1	75.9	73.4	66.8	62.9	57.3	(30)
(31)		5%	WB	48.6	50.4	52.6	55.2	58.4	59.7	63.7	63.4	59.3	56.2	53.0	49.9	(31)
(32)			MCD B	55.8	59.1	62.4	67.3	73.5	79.0	77.7	76.0	73.3	66.7	62.1	56.4	(32)
(33)		10%	WB	47.3	48.9	50.9	53.7	57.1	58.4	62.5	62.2	58.0	54.6	51.7	48.4	(33)
(34)			MCD B	55.3	58.6	63.0	69.4	74.5	79.8	77.9	76.5	73.5	67.6	61.6	56.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.8	14.2	14.1	13.6	14.5	14.0	13.0	12.7	13.2	13.6	13.6	13.7	(35)
(36)			MCDBR	16.1	16.1	15.5	14.6	15.7	14.6	13.8	13.4	13.8	14.3	14.6	16.0	(36)
(37)			MCWBR	10.9	9.9	7.7	6.4	6.4	6.5	6.8	7.2	7.5	6.9	7.8	10.1	(37)
(38)		5% WB	MCD BR	12.9	13.1	12.5	12.6	14.0	13.4	12.7	12.1	12.9	12.3	12.7	11.7	(38)
(39)			MCWBR	10.2	9.8	8.5	7.8	7.6	9.1	8.3	8.0	8.5	7.4	8.2	8.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.306	0.335	0.377	0.422	0.446	0.503	0.569	0.522	0.435	0.354	0.324	0.305	(40)	
(41)		taud	2.475	2.357	2.184	2.065	1.990	1.857	1.734	1.838	2.045	2.301	2.424	2.470	(41)	
(42)		Ebn at Noon	306	302	292	280	271	254	238	250	271	292	297	302	(42)	
(43)		Edh at Noon	31	38	47	54	58	66	74	67	53	39	33	30	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1486	1738	2012	2220	2405	2351	2192	2148	2037	1858	1551	1407	(44)	
(45)		RadStd	65	76	107	117	51	84	89	53	39	52	41	63	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (13 neighbors)		+0.76	+1.22	N/A	N/A	+0.50	+0.54	N/A	N/A	+173	+176	

Nomenclature: See separate page